BUSINESS ANALYSIS PROJECT

Sales Analytics &

Business Intelligence Project

An SQL & Power BI Based Business Analytics Project

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1. Project Overview & Objective

This project is an end-to-end sales analytics solution developed from a Kaggle sales dataset. The project involved preparing the source data for structured analysis, loading it into PostgreSQL, organizing the data into a relational structure, performing business-focused analysis using SQL, and presenting the results through a Power BI dashboard.

The objective was to demonstrate how raw transactional data can be transformed into a structured analytical solution that supports business performance evaluation and decision-making.

> **SCOPE NOTE**
> The detailed database creation SQL, analysis SQL, and Power BI dashboard files are provided separately as project attachments. This document focuses on the business context, project approach, data structure, analytical outcomes, and overall project implementation rather than reproducing those files.

Guiding Business Questions

The project was designed to answer key business questions around sales performance, including:

SL No.	Business Question
1	How is revenue distributed across regions?
2	Which products contribute the most revenue?
3	Which suppliers contribute the most sales?
4	How does revenue change over time?
5	Which customers generate the highest revenue?
6	Which product categories generate the highest profit?

2. Source Dataset Profile

The initial dataset was sourced from **Kaggle** and was provided as a flat CSV dataset.

The dataset was modified before database implementation to make it more suitable for a relational database structure and analytical use.

Final Dataset Profile

Attribute	Value
Records	1,000
Columns	18
Order Records	1,000
Categories	5
Regions	4
Cities	7
Customer Segments	3
Suppliers	6
Payment Methods	6
Analysis Period	February 2024 – February 2025

3. Data Preparation

The original flat-file structure contained information belonging to several business entities within the same dataset.

The dataset was therefore prepared around distinct business entities and identifiers such as:

`customer_id` · `product_id` · `order_id` · `supplier_id`

The resulting structure separated customer, supplier, product, order, and order-item information so that relationships between these entities could be established within PostgreSQL.

This preparation created the foundation for a more structured and reusable analytical database rather than relying solely on a flat CSV file.

4. Database Architecture

The prepared source dataset was first imported into PostgreSQL as the raw_sales table.

From this source layer, the database was organized into five analytical tables:

Table	Purpose
`customers`	Customer segment and geographical information
`suppliers`	Supplier master information
`products`	Product, category, and supplier information
`orders`	Order, customer, date, and payment information
`order_items`	Product-level transactional and financial information

Entity Relationship Flow

The resulting relationship structure follows the business flow:

Customer	→	Order	→	Order Item	→	Product	→	Supplier

5. Analytical Approach

The analysis was structured around six core business questions rather than simply exploring the dataset without a defined objective.

Revenue Performance

Revenue was assessed across geographical regions and over time to identify stronger and weaker areas of sales performance.

Product Performance

Products were ranked by revenue to identify the strongest contributors to overall sales.

Supplier Performance

Supplier-level revenue was assessed to compare the contribution of the available suppliers.

Customer Performance

Customer revenue was evaluated at the city level to identify the highest-value customers and locations.

Category Profitability

Profit was aggregated by product category to assess which categories contributed most to overall profitability.

6. Key Findings

Overall Performance

The completed analysis produced the following overall results:

₹75.21M	₹15.04M	3,097	1,000	~20%
Total Revenue	Total Profit	Total Quantity	Total Orders	Profit Margin

Regional Performance

Regional revenue was distributed as follows:

Region	Revenue
North	₹20.29M
South	₹19.86M
West	₹18.39M
East	₹16.67M

Product Performance

The highest-revenue products were:

Rank	Product	Revenue
1	Educational Book	₹4.52M
2	Laptop	₹4.13M
3	Table Lamp	₹3.99M
4	Headphones	₹3.72M
5	Jeans	₹3.69M

Supplier Performance

Supplier revenue was distributed as follows:

Supplier	Revenue
Uniqlo	₹15.11M
Kodansha	₹14.79M
Shiseido	₹14.68M
Zojirushi	₹13.33M
Sony	₹9.25M

Supplier	Revenue
Panasonic	₹8.06M

Monthly Performance

Revenue was analyzed on a monthly basis across the available period.

The trend demonstrates noticeable variation rather than a uniform sales pattern. July 2024 recorded approximately **₹7.64M**, while February 2025 recorded approximately **₹1.51M**.

Category Profitability

Category	Profit
Electronics	₹3.46M
Clothing	₹3.02M
Books	₹2.96M
Beauty	₹2.94M
Home & Kitchen	₹2.67M

Customer and City Performance

> **HIGHLIGHT**
> The analysis identified **Kolkata** as the highest-revenue city, generating approximately **₹16.67M**.

The customer analysis also incorporated customer segmentation, allowing performance to be viewed across **Wholesale, Online, and Retail** customers.

Revenue across these segments was relatively balanced:

Customer Segment	Revenue
Wholesale	₹25.73M
Online	₹25.17M
Retail	₹24.31M

7. Power BI Reporting Solution

The SQL analysis was subsequently brought into Power BI to create an interactive reporting layer.

The dashboard was structured into three business-focused areas:

Executive Sales Overview

Provides an overall view of revenue, profit, quantity, orders, margin, monthly performance, regional performance, category performance, and customer segments.

Product & Supplier Analysis

Provides deeper analysis of product, category, and supplier performance, including revenue, quantity, profit, and product-level comparisons.

Customer & Regional Analysis

Provides city, regional, customer-segment, and payment-method perspectives on sales performance.

8. Business Interpretation

The analysis establishes several meaningful areas for management attention.

North is the strongest region by revenue, while East is comparatively weaker. Electronics is the leading profit-generating category, making it a particularly important contributor to overall profitability. Educational Book is the highest-revenue individual product, while **Uniqlo** is the highest-revenue supplier in the dataset.

At the customer level, Kolkata stands out as the strongest city by revenue. At the same time, the relatively even contribution of Wholesale, Online, and Retail segments indicates a diversified customer base rather than dependence on one segment.

These findings provide a starting point for deeper investigation into regional opportunity, product strategy, supplier management, and customer development.

9. Project Outputs & Value

Project Outputs

The completed project produces four primary outputs:

SL No.	Output
1	Prepared analytical dataset
2	PostgreSQL relational database
3	SQL business analysis queries
4	Power BI interactive dashboard

Project Value

The project demonstrates an end-to-end process of converting raw business data into structured analysis and management-oriented reporting.

The main value of the solution is the integration of three stages:

Data	→	Analysis	→	Decision Support

Rather than treating SQL and Power BI as isolated technical exercises, the project connects database structuring with defined business questions and presents the resulting analysis in a form suitable for business review.

10. Conclusion

This project establishes a complete analytical workflow from a Kaggle-based raw dataset through data preparation, PostgreSQL database structuring, SQL-based business analysis, and Power BI reporting.

The completed solution provides visibility into **revenue, profit, products, categories, suppliers, customers, regions, cities, and customer segments**, while maintaining a clear separation between the underlying data layer, analytical layer, and reporting layer.

The project demonstrates the practical application of **SQL and Power BI within a business analysis context**, with the emphasis on converting transactional data into structured information and business-relevant insights.

11. Project Repository Contents

The documentation is supported by the corresponding implementation files:

SL No.	File / Component
1	Dataset_Modified
2	Database_Creation
3	Analysis_Queries
4	Excecutive Dashboard